

More Americans, Fewer Storm Deaths Per Person

What 86 years of U.S. weather data say about adaptation

Deaths from tornadoes, floods and hurricanes held roughly steady while the population more than doubled. Heat is the exception, and how heat deaths are counted changes the answer. By **Omer Okutan** · Fall 2026

262 → 195

tornado, flood & hurricane deaths per year, 1940s vs. 2010s

1.88 → 0.61

the same deaths per million residents per year (−68%)

+2.9 million

net domestic movers into the hottest 40% of counties, 2020–25

555 vs. ~120,000

NWS-counted heat deaths (2023) vs. estimated U.S. deaths from heat + cold per year

1 STORM DEATHS VS. POPULATION, 1940–2025

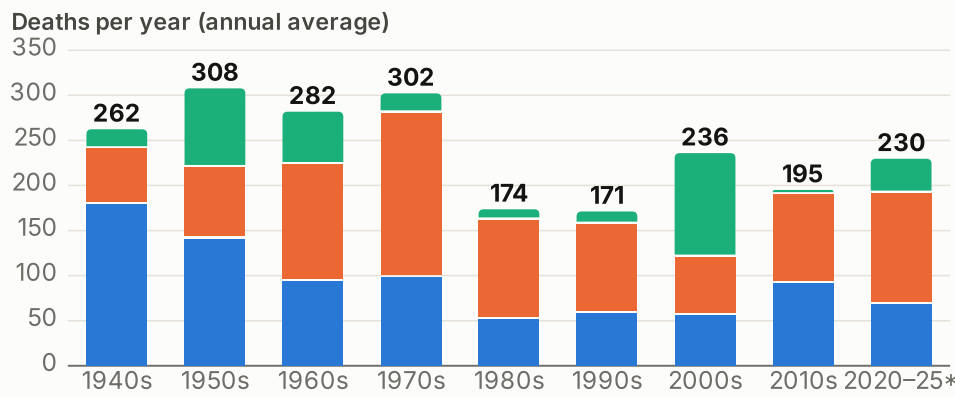
Storm deaths per person fell by about two-thirds

Tornadoes killed about as many Americans in 2011 (553) as in 1953 (515), but the population nearly doubled (160M → 312M). Over 1940–2025, tornadoes, floods and hurricanes killed about **240 people a year**, with no clear trend in the raw count.^{1,3}

Annual deaths: no clear trend

Annual-average U.S. deaths by decade

Tornado Flood Hurricane (wind only)

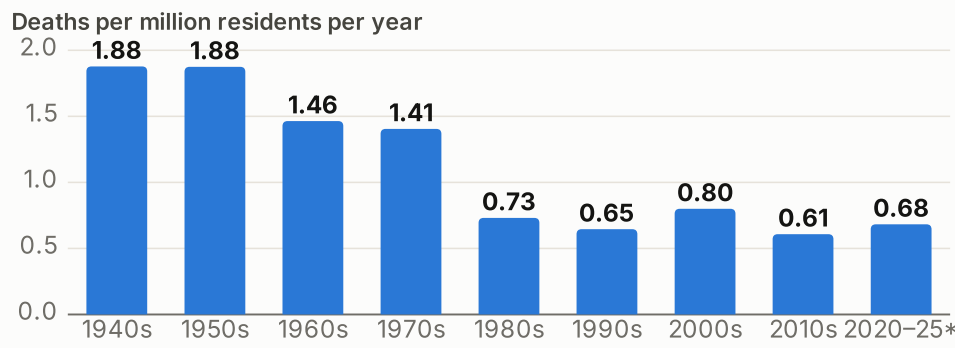


*2020–25 = six years; 2025 preliminary. The 2000s include Hurricane Katrina (1,016 NWS hurricane deaths in 2005). Source: NWS.¹

Death intensity = annual deaths ÷ average population × 1,000,000. It fell from **1.88 to 0.61** deaths per million residents per year.

Deaths per million residents

Tornado + flood + hurricane deaths per million U.S. residents per year



Population: BEA midyear resident series, averaged by decade.³

Caveats: NWS “hurricane” deaths count wind only (surge and rain-flood deaths are filed as Flood), and a single storm can move a whole decade. The CDC, not the NWS, is the official cause-of-death source.²

Possible reasons (not tested here): higher incomes, better warnings, stronger building codes. One has been measured: a very hot day’s mortality effect fell ~80% from 1900–59 to 1960–2004, and home AC explains nearly all of it.¹³

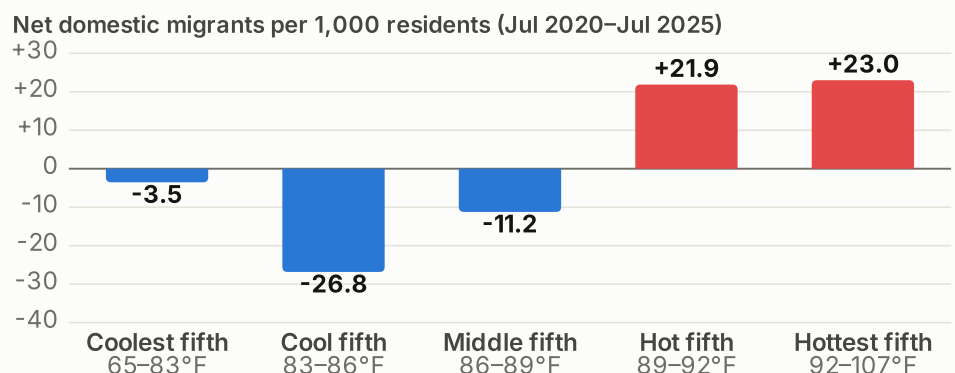
2 WHERE AMERICANS ARE MOVING, 2020–2025

Movers are heading to hotter counties and milder winters

I split 3,098 counties (98% of the population) into five equal-population groups by average July high and summed Census **net domestic migration** (excludes births, deaths and immigration).^{4,5}

Net domestic migration by county summer heat

Net domestic migrants per 1,000 residents, July 2020–July 2025



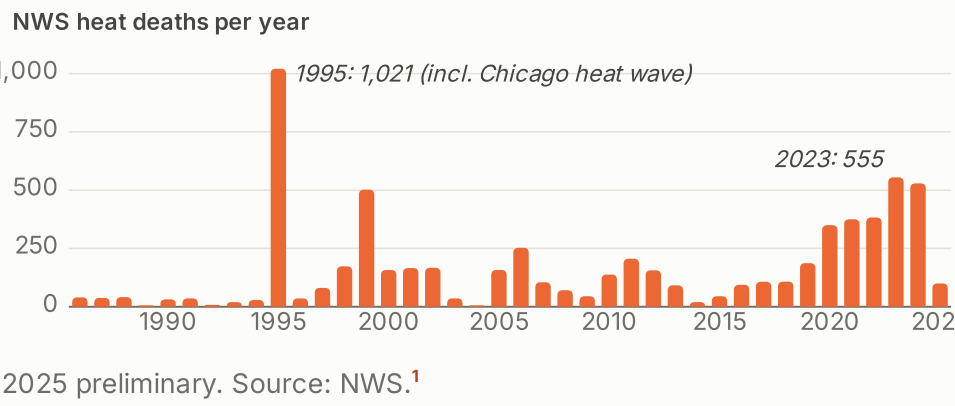
Contiguous U.S. (excludes CT). Heat = 1991–2020 average July high (NOAA nClimDiv). Census Vintage 2025.^{4,5}

The hottest two fifths gained **~2.9M** net movers. **Scale matters:** hot urban cores like Dallas County (−191k) and Miami-Dade (−271k) lost residents, many to nearby hot suburbs, and the mildest-winter fifth gained 1.3M. Migration raises heat exposure but lowers cold exposure, which may help health: Northeast-to-Southwest moves account for 4–7% of U.S. life-expectancy gains over 30 years.⁷

3 HEAT DEATHS, 1986–2025

Reported heat deaths rose sharply in the 2020s

NWS heat deaths per year, 1986–2025



2025 preliminary. Source: NWS.¹

NWS heat deaths averaged ~115/yr in the 2000s–2010s and **382/yr in 2020–25**, more than tornadoes, floods and hurricanes combined (230/yr). The NWS undercounts: in 2023 **Maricopa County alone recorded 645** vs. the NWS’s 555 for the U.S. CDC death certificates show 2,325 and a sharp rise since 2016.^{10,9} In Chicago’s 1995 heat wave, people with working AC had one-third the odds of dying.¹¹

4 COUNTED VS. ESTIMATED DEATHS

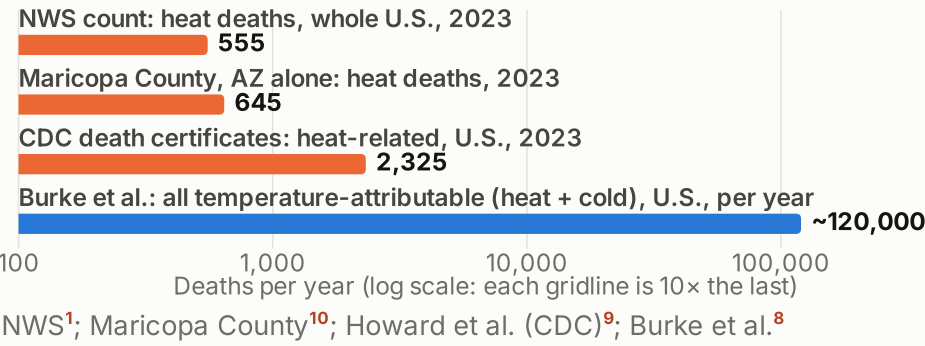
“Temperature deaths” can mean 555 or 120,000

Four ways to count a “temperature death”

Annual U.S. deaths, log scale

Heat, counted case by case

Heat + cold, statistically estimated



100 1,000 10,000 100,000 Deaths per year (log scale: each gridline is 10 × the last)

NWS¹; Maricopa County¹⁰; Howard et al. (CDC)⁹; Burke et al.⁸

Burke et al. (2025) attribute ~5% of U.S. and ~12% of EU deaths to non-optimal temperatures: ~120,000 Americans a year.⁸ A different measure from NWS counts:

- Counted** (NWS, CDC): a specific death is linked to weather by a coroner, doctor or forecaster.
- Estimated** (Burke): extra deaths vs. every day at the lowest-risk temperature (~20–25°C), largely heart and lung deaths. Only 0.058% of U.S. death records (1968–2019) mention temperature.
- Cold deaths outnumber heat deaths at least 5 to 1** in Burke’s sample, mostly on moderately cold days.

If 5–12% applied to all 61.7M deaths worldwide: **3.1–7.4 million a year.**²⁰

5 COSTS OF ADAPTING

Relocating coastal buildings would cost ~0.01% of GDP a year

Risk = hazard × exposure. Course figures from Nordhaus: ~\$600B of structures sit below 10 m in Atlantic hurricane zones.¹⁵ Rebuild each one on safer ground as it wears out:

\$600B (assumed 2019 \$) × CPI → 2025 \$	≈ \$756B
Assumed 50-yr life → 2% replaced/yr	≈ \$15.1B
Assumed extra 25% to rebuild elsewhere	≈ \$3.8B/yr
Share of 2025 GDP (\$30.8T) ¹⁷	≈ 0.012%

Even with a 30-year life and 100% relocation cost: ~0.08% of GDP. Yet Nordhaus saw no sign of such relocation.¹⁶ For temperature, the best evidence favors non-climate policy: community health centers cut U.S. heat deaths 14%.⁸

What could make these conclusions wrong

Counting rules

Surge deaths are filed under Flood, and heat deaths are undercounted, so reporting shapes both trends.

Averages hide people

A falling death rate doesn’t tell us whether the *next* dollar of protection is worth spending. That is a marginal question.

Adaptation can stall

The U.S. temperature burden was the same share of deaths in the 2010s as in the 1980s, partly because the population is older.⁸

Cold vs. winter

Winter also brings flu and indoor crowding. Separating the effect of a cold *day* from winter itself needs more evidence.

A big number is not a verdict. Neither is a small one.

Americans got far better at surviving storms, even as they moved toward heat. Most temperature deaths happen quietly, on ordinary cold days. The useful question is where the next dollar of protection saves the most lives, and the evidence points to exposure and vulnerability (where we build, who can cool or heat their home, access to care) as much as the weather itself.



Full story, data & methods: fishman2341.github.io/econ238-portfolio/week-four/weather-mortality

1. NWS, 81-Year List of Severe Weather Fatalities 1940–2025 (weather.gov/hazstat)
2. NWS, Weather Related Fatality & Injury Statistics notes (weather.gov/hazstat)
3. BEA population, FRED B230RC0A052NBEA
4. U.S. Census Bureau, County Components of Change, Vintage 2025
5. NOAA NCEI nClimDiv county temperature & degree days
6. U.S. Census Bureau, Mean Center of Population 1790–2020
7. Deschênes & Moretti (2009), Review of Economics & Statistics 91(4)
8. Burke et al. (2025), NBER Working Paper 34313

9. Howard et al. (2024), JAMA, heat deaths 1999–2023
10. Maricopa County Public Health, heat deaths 2023–24 (maricopa.gov)
11. Semenza et al. (1996), NEJM 335(2), Chicago 1995
12. Naughton et al. (2002), Am. J. Prev. Med. 22(4), Chicago 1999
13. Barreca et al. (2016), Journal of Political Economy 124(1)
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17. BEA GDP & BLS CPI-U via FRED (GDPa, CPIAUCSL)
18. Zhao et al. (2021), Lancet Planetary Health
19. CDC/NCHS Data Brief 548, Mortality in the U.S., 2024
20. UN World Population Prospects 2024 (via Our World in Data)
21. Ballester et al. (2023), Nature Medicine
Full links on the webpage (QR code).